Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Powers of i**

Evaluate each power of i, using $i^2 = -1$ and the fact that powers of i repeat every 4:

a $i^3 = -i$ **b** $i^4 = 1$ **c** $i^7 = -i$ **d** $i^{22} = -1$

(a) $= -i$; (b) $= 1$; (c) $= -i$; (d) $= -1$

2 **[Foundational]** • **Target: Multiplication**

Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.

$= 25 - (-4) + 0i = 29$. It is a product of conjugates $z\bar{z} = |z|^2$ (Theorem 1.1), which is always real.

3 **[Foundational]** • **Target: Conjugate and modulus**

Show that $z + \bar{z}$ is always real for $z = a + bi$.

$z + \bar{z} = (a + bi) + (a - bi) = 2a$, which is real since a is real.

4 **[Foundational]** • **Target: Division**

Simplify $\frac{3 + 4i}{1 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

Multiply by $\frac{1 + 2i}{1 + 2i}$: numerator $= (3)(1) - (4)(2) + [(3)(2) + (4)(1)]i = -5 + 10i$; denominator $= 1^2 + 2^2 = 5$. So the quotient is $-1 + 2i$.

5 **[Foundational]** • **Target: Quadratic equations with complex roots**

Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

Discriminant $= 36 - 52 = -16$, so $\sqrt{-16} = 4i$. $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$.

6 **[Foundational]** • **Target: Quadratic equations with complex roots**

Solve the quadratic equation $2x^2 - 6x + 9 = 0$ over the complex numbers, and state the sum and product of the roots.

Step 1: Compute the discriminant: $\Delta = b^2 - 4ac = (-6)^2 - 4(2)(9) = 36 - 72 = -36$. Step 2: Apply the quadratic formula: $x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{6 \pm \sqrt{-36}}{2(2)} = \frac{6 \pm 6i}{4} = \mathbf{\frac{3}{2} \pm \frac{3}{2}i}$. Step 3: Verify using Vieta's formulas: Sum of roots: $\left(\frac{3}{2} + \frac{3}{2}i\right) + \left(\frac{3}{2} - \frac{3}{2}i\right) = 3 = -\frac{b}{a} = -\frac{-6}{2} = 3$ ✓. Product of roots.

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7 [Foundational] • Target: Powers of i

The polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ has $(x - 2)$ as a factor. (a) Find the value of k. (b) Factor $P(x)$ completely over the real numbers and list all its roots.

Part (a): Find k via the Factor Theorem: By Theorem 4.1, $(x - 2)$ is a factor if and only if $P(2) = 0$: $P(2) = 2(2)^3 + k(2)^2 - 7(2) + 6 = 0$ $2(8) + 4k - 14 + 6 = 16 + 4k - 8 = 4k + 8 = 0 \Rightarrow 4k = -8 \Rightarrow \mathbf{k = -2}$. Part (b): Divide and factor completely: Polynomial division of $2x^3 - 2x^2 - 7x + 6$ by $x - 2$: $2x^3 - 2x^2 - 7x + 6 = (x - 2)(2x^2 + 2x - 3)$. Solve $2x^2 + 2x - 3 = 0$ using the quadratic formula.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Addition and subtraction

Express $z = \frac{4 + 7i}{1 - 2i}$ in standard form $a + bi$, and compute its modulus squared $|z|^2$.

Step 1: Multiply numerator and denominator by the conjugate $1 + 2i$: $z = \frac{(4 + 7i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{4(1) - 7(2) + [4(2) + 7(1)]i}{1^2 + (-2)^2}$ Step 2: Simplify numerator and denominator: $z = \frac{4 - 14 + (8 + 7)i}{1 + 4} = \frac{-10 + 15i}{5} = \mathbf{-2 + 3i}$. Step 3: Compute modulus squared: $|z|^2 = (-2)^2 + 3^2 = 4 + 9 = \mathbf{13}$. (Verification via quotient property: $|z|^2 = \frac{|4 + 7i|^2}{|1 - 2i|^2} = \frac{4^2 + 7^2}{1^2 + (-2)^2} = \frac{16 + 49}{1 + 4} = \frac{65}{5} = 13$ ✓).

9 [Intermediate] • Target: Multiplication

Find the Cartesian equation in terms of x and y of the locus of complex numbers $z = x + yi$ satisfying the equation $|z - 3| = |z + 3i|$, and describe its geometric shape.

Step 1: Substitute $z = x + yi$: $|(x - 3) + yi| = |x + (y + 3)i|$ Step 2: Square both sides using modulus definition: $(x - 3)^2 + y^2 = x^2 + (y + 3)^2$ Step 3: Expand the squares: $x^2 - 6x + 9 + y^2 = x^2 + y^2 + 6y + 9$ Step 4: Cancel common terms $x^2, y^2, 9$: $-6x = 6y \Rightarrow \mathbf{y = -x}$. Geometric description: This locus is a straight line through the origin with slope -1, representing the perpendicular bisector of the segment connecting $(3, 0)$ and $(0, -3)$.

10 [Intermediate] • Target: Conjugate and modulus

Compute $(1 - i)^{10}$ in exact rectangular form $a + bi$ using polar representation.

Step 1: Convert $1 - i$ to polar form: Modulus: $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. Argument: $a = 1 > 0, b = -1 Step 2 : Apply De Moivre's Theorem : (1 - i)^{10} = (\sqrt{2})^{10} \left[\cos\left(-\frac{10\pi}{4}\right) + i\sin\left(-\frac{10\pi}{4}\right)\right] = 2^5 \left[\cos\left(-\frac{5\pi}{2}\right) + i\sin\left(-\frac{5\pi}{2}\right)\right]. x$. $\cos\left(-\frac{\pi}{2}\right) = 0$ and $\sin\left(-\frac{\pi}{2}\right) = -1$. $(1 - i)^{10} = 32(0 - i) = \mathbf{-32i}$.

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11 **[Intermediate]** • **Target: Division**

Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.

Step 1: Compute modulus r: $r = \sqrt{(-2\sqrt{3})^2 + 2^2} = \sqrt{12+4} = \sqrt{16} = 4$. Step 2: Determine argument θ: Since $a = -2\sqrt{3} < 0$ and $b = 2 > 0$, z lies in Quadrant II. Reference angle: $\tan\alpha = \left|\frac{2}{-2\sqrt{3}}\right| = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$ (or 30°). Quadrant II angle: $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ (or 150°). Step 3: Write polar form.

12 **[Intermediate]** • **Target: Quadratic equations with complex roots**

Consider the function $f(x) = \sqrt{2x-5} + 3$. (a) Find the domain and range of $f(x)$. (b) Find the explicit equation for the inverse function $f^{-1}(x)$, including its restricted domain.

Part (a): Domain and Range of f: 1. Radicand non-negative: $2x - 5 \geq 0 \Rightarrow 2x \geq 5 \Rightarrow \text{dom}\mathbf{(f) = \left[\frac{5}{2}, \infty\right)}$. 2. Since $\sqrt{2x-5} \geq 0$, $f(x) = \sqrt{2x-5} + 3 \geq 3 \Rightarrow \text{range}\mathbf{(f) = [3, \infty)}$. Part (b): Derive $f^{-1}(x)$: 1. Set $y = \sqrt{2x-5} + 3$ and swap x and y: $x = \sqrt{2y-5} + 3$ 2. Isolate the square root: $x - 3 = \sqrt{2y-5}$ 3. Square both sides (valid for $x \geq 3$).

13 **[Intermediate]** • **Target: Powers of i**

Solve the logarithmic equation for x: $\log_3(x+2) + \log_3(x-4) = 3$

Step 1: Determine domain restrictions: $x + 2 > 0 \Rightarrow x > -2$ and $x - 4 > 0 \Rightarrow x > 4$. Both hold simultaneously when $\mathbf{x > 4}$. Step 2: Combine logarithms using the product property: $\log_3((x+2)(x-4)) = 3$ Step 3: Convert to exponential form: $(x+2)(x-4) = 3^3 = 27$ $x^2 - 2x - 8 = 27 \Rightarrow x^2 - 2x - 35 = 0$. Step 4: Factor the quadratic: $(x-7)(x+5) = 0 \Rightarrow x = 7 \quad \text{or} \quad x = -5$. Step 5: Apply domain check: $x = -5$ produces $\log_3(-3)$, which is.

14 **[Intermediate]** • **Target: Addition and subtraction**

Solve the rational equation for all real values of x: $\frac{x}{x-2} + \frac{1}{x-4} = \frac{2}{(x-2)(x-4)}$

Step 1: Identify domain restrictions: Denominators cannot be zero: $x \neq 2$ and $x \neq 4$. Step 2: Clear denominators by multiplying by $(x-2)(x-4)$: $x(x-4) + 1(x-2) = 2$ $x^2 - 4x + x - 2 = 2 \Rightarrow x^2 - 3x - 4 = 0$. Step 3: Factor the quadratic equation: $(x-4)(x+1) = 0 \Rightarrow x = 4 \quad \text{or} \quad x = -1$. Step 4: Check for extraneous solutions: $x = 4$ violates the domain restriction $x \neq 4$ (it produces division by zero), and must be rejected. $x = -1$ is valid.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Multiplication** • **Cross-Lesson: 6.01 Limits**

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

Part 1: Exact value of $\cos(75^\circ)$: Express $75^\circ = 45^\circ + 30^\circ$ and apply the cosine sum identity: $\cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}-\sqrt{2}}{4} = \frac{\sqrt{\mathbf{6}}-\sqrt{\mathbf{2}}}{\mathbf{4}}$. Part 2: Evaluate $\tan(2\theta)$: Apply the double-angle identity for tangent.

16 **[Challenging]** • **Target: Conjugate and modulus** • **Cross-Lesson: 2.01 Matrices & Vectors**

Evaluate the limit using algebraic conjugate rationalization: $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$

Step 1: Check form upon direct substitution: $\frac{\sqrt{4}-2}{4-4} = \frac{0}{0}$ (indeterminate form). Step 2: Multiply by the conjugate of the numerator: $\lim_{x\to 4} \frac{(\sqrt{x}-2)(\sqrt{x}+2)}{(x-4)(\sqrt{x}+2)} = \lim_{x\to 4} \frac{(\sqrt{x})^2 - 2^2}{(x-4)(\sqrt{x}+2)} = \lim_{x\to 4} \frac{x-4}{(x-4)(\sqrt{x}+2)}$. Step 3: Cancel the common factor $(x-4)$ for $x \neq 4$.

17 **[Challenging]** • **Target: Division** • **Cross-Lesson: 3.01 Sequences & Series**

Find all complex solutions to the cubic equation $z^3 + 8 = 0$.

Step 1: Rewrite as a difference of cubes: $z^3 - (-8) = 0 \Rightarrow z^3 + 2^3 = (z+2)(z^2 - 2z + 4) = 0$. Step 2: Real root: $z + 2 = 0 \Rightarrow \mathbf{z_1 = -2}$. Step 3: Complex roots via quadratic formula: $z^2 - 2z + 4 = 0 \Rightarrow z = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(4)}}{2} = \frac{2 \pm \sqrt{4-16}}{2} = \frac{2 \pm \sqrt{-12}}{2} = \frac{2 \pm 2i\sqrt{3}}{2} = \mathbf{1 \pm i\sqrt{3}}$. Thus the three roots are $\mathbf{-2,\quad 1 + i\sqrt{3},\quad 1 - i\sqrt{3}}$.

18 **[Challenging]** • **Target: Quadratic equations with complex roots** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

Identify the geometric shape in the complex plane described by the equation $|z-4| + |z+4| = 10$, and calculate the length of its minor axis.

Step 1: Recognize definition of an ellipse: The sum of distances from z to the foci $F_1(4,0)$ and $F_2(-4,0)$ is constant: $2a = 10 \Rightarrow a = 5$. Step 2: Compute focal distance c: Foci at $(\pm 4, 0) \Rightarrow c = 4$. Step 3: Relate semi-axes: $b^2 = a^2 - c^2 = 5^2 - 4^2 = 25 - 16 = 9 \Rightarrow b = 3$. Conclusion: The locus is an ellipse with major axis $2a = 10$ and minor axis $2b = 2(3) = \mathbf{6}$.

19 **[Challenging]** • **Target: Powers of i** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Simplify the complex quotient into standard form $a + bi$: $\frac{3+4i}{1+2i}$

Step 1: Multiply numerator and denominator by the conjugate $1 - 2i$: $\frac{3+4i}{1+2i} = \frac{(3+4i)(1-2i)}{(1+2i)(1-2i)}$ Step 2: Expand numerator and denominator: Numerator: $3(1) - 4(-2) + [3(-2) + 4(1)]i = 3 + 8 + (-6+4)i = 11 - 2i$. Denominator: $1^2 + 2^2 = 1 + 4 = 5$. Step 3: Separate real and imaginary parts: $\frac{11-2i}{5} = \mathbf{\frac{11}{5} - \frac{2}{5}i}$.

20 **[Challenging]** • **Target: Addition and subtraction** • **Cross-Lesson: 6.01 Limits**

If r_1 and r_2 are the roots of $3x^2 - 8x + 5 = 0$, find the exact value of $\frac{1}{r_1} + \frac{1}{r_2}$ without solving for the roots individually.

Step 1: Combine into common denominator: $\frac{1}{r_1} + \frac{1}{r_2} = \frac{r_1 + r_2}{r_1 r_2}$ Step 2: Apply Vieta's formulas: Sum of roots: $r_1 + r_2 = -\frac{b}{a} = -\frac{-8}{3} = \frac{8}{3}$. Product of roots: $r_1 r_2 = \frac{c}{a} = \frac{5}{3}$. Step 3: Divide: $\frac{r_1 + r_2}{r_1 r_2} = \frac{8/3}{5/3} = \mathbf{\frac{8}{5}}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Powers of i

Evaluate i^{130}, using the fact that powers of i repeat every 4.

a. i

b. $-i$

c. 1

d. -1 ✓

$130 = 4(32) + 2$, so $i^{130} = (i^4)^{32} \cdot i^2 = 1 \cdot -1 = -1$.

22 [Foundational] • Target: Addition and subtraction

Find the product of the roots of $x^2 + 4x + 13 = 0$.

a. **13** ✓

b. 15

c. 4

d. -13

By Vieta's formulas, the product of the roots of $x^2 + Bx + C = 0$ is C; here that's 13 (the imaginary parts of the conjugate pair cancel in the sum, and multiply to a real number in the product).

23 [Foundational] • Target: Multiplication

Find the sum of the two roots of $x^2 - 6x + 13 = 0$.

a. 6 ✓

b. 3

c. 13

d. $4i$

By Vieta's formulas the sum of the roots of $x^2 - 6x + 13 = 0$ is $-(-6)/1 = 6$ (the complex parts of the conjugate pair $3 \pm 2i$ cancel).

24 [Foundational] • Target: Conjugate and modulus

What is the complex conjugate of $5 - 2i$?

a. $5 + 2i$ ✓

b. $-5 + 2i$

c. $5 - 2i$

d. $-5 - 2i$

The conjugate flips the sign of the imaginary part only: $5 - 2i \to 5 + 2i$.

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25 [Foundational] • Target: Division

For which value of k does $x^2 + kx + 9 = 0$ have complex (non-real) roots?

a. $k = 5$ ✓

b. $k = 7$

c. $k = 6$

d. $k = -8$

Complex roots require $k^2 - 36 < 0$, i.e. $-6 < k < 6$. Only $k = 5$ is in that range.

26 [Foundational] • Target: Quadratic equations with complex roots

Find the complex conjugate of $7 - 3i$.

a. **7+3i** ✓

b. -7-3i

c. -7+3i

d. 7-3i

Flip the sign of the imaginary part only: $7 - 3i \to 7 + 3i$.

27 [Foundational] • Target: Powers of i

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?

a. -5

b. -3

c. 1

d. 5 ✓

Step 1: Find the equation of $f(x)$. Using the vertex form of a parabola with vertex $(h, k_v) = (2, -1)$: $f(x) = a(x - 2)^2 - 1$ Since the graph passes through the y-intercept $(0, 3)$, substitute $x = 0$ and $y = 3$: $3 = a(0 - 2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$ Thus, $f(x) = (x - 2)^2 - 1$. Step 2: Apply the condition that g passes through $(0, 0)$. Since $g(0) = 0$ and $g(x) = f(x + k) - 8$: $g(0) = f(0 + k) - 8 = f(k) - 8 = 0 \Rightarrow f(k) = 8$ Step 3: Solve for k..

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28 [Foundational] • Target: Addition and subtraction

Let z be a complex number satisfying the equation $|z-12|+|z+12|=26$. What is the maximum possible value of $|z-5i|$?

a. $\frac{169}{12}$ ✓

b. $\sqrt{194}$

c. 18

d. 10

1. **Identify the geometric locus:** The equation $|z-12|+|z+12|=26$ represents an ellipse in the complex plane with foci at $z_1=12$ and $z_2=-12$ along the real axis. - The focal distance is $2c=24 \Rightarrow c=12$. - The major axis length is $2a=26 \Rightarrow a=13$. - The semi-minor axis length is $b=\sqrt{a^2-c^2}=\sqrt{13^2-12^2}=\sqrt{25}=5$. In Cartesian coordinates $z=x+iy$, the equation of this ellipse is.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiplication

In the complex plane, let z be a complex number that satisfies the equation $|z-4|+|z+4|=10$. What is the maximum possible value of $|z-3i|$?

a. $\frac{25}{4}$ ✓

b. 6

c. $\sqrt{34}$

d. $\frac{13}{2}$

1. **Identify the locus in the complex plane:** The equation $|z-4|+|z+4|=10$ defines the set of all points $z=x+yi$ whose sum of distances to the fixed points $(4,0)$ and $(-4,0)$ is constant and equal to 10. This is the definition of an ellipse with foci at $(\pm 4,0)$. - Major axis length: $2a=10 \Rightarrow a=5$. - Focal distance: $c=4$. - Semi-minor axis: $b=\sqrt{a^2-c^2}=\sqrt{25-16}=3$. Therefore, the equation of the ellipse in Cartesian coordinates is.

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30 [Intermediate] • Target: Conjugate and modulus

What is the minimum value of the function $f(x) = \sqrt{x^2 - 4x + 13} + \sqrt{x^2 - 14x + 130}$ for all real numbers x?

a. 12

b. $4\sqrt{10}$

c. **13** ✓

d. $\sqrt{170}$

To find the minimum value of $f(x)$, synthesize the algebraic expressions with coordinate geometry by completing the square under each radical: 1. Rewrite the radicands: $x^2 - 4x + 13 = (x-2)^2 + 9 = (x-2)^2 + 3^2$ $x^2 - 14x + 130 = (x-7)^2 + 81 = (x-7)^2 + 9^2$ 2. Interpret geometrically using the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$: - $\sqrt{(x-2)^2 + 3^2}$ is the distance from the point $P(x, 0)$ on the x-axis to the fixed point $A(2, 3)$..

31 [Intermediate] • Target: Division

For what real value of k is the complex expression $\frac{k+8i}{3-i} + \frac{2-ki}{3+i}$ a real number?

a. **11** ✓

b. 1

c. 6

d. -11

To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates. The common denominator of the two fractions is: $(3-i)(3+i) = 3^2 - i^2 = 9 - (-1) = 10$ Rewrite each term over this denominator: 1. For the first term: $\frac{(k+8i)(3+i)}{(3-i)(3+i)} = \frac{3k + ki + 24i + 8i^2}{10} = \frac{(3k-8) + (k+24)i}{10}$ 2. For the second term.

32 [Intermediate] • Target: Quadratic equations with complex roots

In the complex plane, the complex numbers z and w satisfy $|z| = 5$, $|w| = 7$, and $|z + w| = 10$. What is the value of $|z - w|$?

a. $4\sqrt{3}$ ✓

b. $2\sqrt{6}$

c. $4\sqrt{6}$

d. $2\sqrt{37}$

To find $|z - w|$, we use the algebraic properties of the modulus and complex conjugates, which correspond to the parallelogram law in the complex plane. Recall that for any complex number u, $|u|^2 = u\overline{u}$. Expanding the modulus squared of the sum and difference of z and w: 1. $|z + w|^2 = (z + w)(\overline{z} + \overline{w}) = |z|^2 + |w|^2 + (z\overline{w} + \overline{z}w)$ 2. $|z - w|^2 = (z - w)(\overline{z} - \overline{w}) = |z|^2 + |w|^2 - (z\overline{w} + \overline{z}w)$ Adding these two equa.

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33 **[Intermediate]** • **Target: Powers of i**

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation $f(x) + f\left(\frac{x-1}{x}\right) = 1 + x$. What is the value of $f(2)$?

a. $\frac{3}{4}$ ✓

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

To find $f(2)$, substitute strategic values of x to generate a system of equations involving the cyclic transformation $g(x) = \frac{x-1}{x}$. 1. Substitute $x = 2$: $\frac{2-1}{2} = \frac{1}{2}$ $f(2) + f\left(\frac{1}{2}\right) = 1 + 2 = 3 \quad - - - \quad$ (Equation 1) 2. Substitute $x = \frac{1}{2}$: $\frac{\frac{1}{2}-1}{\frac{1}{2}} = \frac{-\frac{1}{2}}{\frac{1}{2}} = -1$ $f\left(\frac{1}{2}\right) + f(-1) = 1 + \frac{1}{2} = \frac{3}{2} \quad - - - \quad$ (Equation 2) 3. Substitute $x = -1$.

34 **[Intermediate]** • **Target: Addition and subtraction**

In the complex number system, let $i^2 = -1$. For what real value of k is the expression $\frac{6+ki}{3+2i}$ a purely imaginary number?

a. **-9** ✓

b. -4

c. 4

d. 9

To determine when the quotient is purely imaginary, multiply the numerator and denominator by the complex conjugate of the denominator, $3 - 2i$: $\frac{6+ki}{3+2i} = \frac{(6+ki)(3-2i)}{(3+2i)(3-2i)}$ Expand the denominator using $i^2 = -1$: $(3 + 2i)(3 - 2i) = 3^2 - (2i)^2 = 9 - 4(-1) = 13$ Expand the numerator: $(6 + ki)(3 - 2i) = 18 - 12i + 3ki - 2ki^2 = 18 + 2k + (3k - 12)i$ Thus, the expression in standard form $a + bi$ is: $\frac{18+2k}{13} + \frac{3k-12}{13}i$ A complex number is purely imag.

35 **[Intermediate]** • **Target: Multiplication**

For the imaginary unit i, where $i^2 = -1$, the expression $\frac{k+6i}{3-2i}$ is equal to a real number for some real constant k. What is the value of k?

a. **-9** ✓

b. -4

c. 4

d. 9

To simplify the quotient of two complex numbers, multiply the numerator and denominator by the complex conjugate of the denominator, $3 + 2i$: $\frac{k+6i}{3-2i} = \frac{(k+6i)(3+2i)}{(3-2i)(3+2i)}$ Expand the denominator: $(3 - 2i)(3 + 2i) = 3^2 - (2i)^2 = 9 - 4(-1) = 9 + 4 = 13$ Expand the numerator: $(k + 6i)(3 + 2i) = 3k + 2ki + 18i + 12i^2 = 3k + 12(-1) + (2k + 18)i = (3k - 12) + (2k + 18)i$ Thus, the expression can be written in standard form $a + bi$ as: $\frac{3k-12}{13} + \frac{2k+18}{13}i$.

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36 [Intermediate] • Target: Conjugate and modulus

For the imaginary unit i, where $i^2 = -1$, which of the following is equivalent to the expression $(5 - 2i)(3 + 4i)$?

a. $23 + 14i$ ✓

b. $7 + 14i$

c. $23 + 26i$

d. $7 - 14i$

To multiply the two complex numbers, apply the distributive property (FOIL method): $(5 - 2i)(3 + 4i) = (5)(3) + (5)(4i) + (-2i)(3) + (-2i)(4i)$ Compute each term: - First: $5 \times 3 = 15$ - Outside: $5 \times 4i = 20i$ - Inside: $-2i \times 3 = -6i$ - Last: $-2i \times 4i = -8i^2$ Substitute $i^2 = -1$: $-8i^2 = -8(-1) = 8$ Now, group and combine the real and imaginary components: $\text{Real part}: \ 15 + 8 = 23$ $\text{Imaginary part}: \ 20i - 6i = 14i$ Thus, the product is $23 + 14i$. - Choice B (.

37 [Intermediate] • Target: Division

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

a. $3 + 4i$ ✓

b. $3 - 10i$

c. $7 + 4i$

d. $3 - 4i$

To subtract complex numbers, distribute the negative sign across the second term and combine like terms (real parts with real parts, and imaginary parts with imaginary parts): $(5 - 3i) - (2 - 7i) = 5 - 3i - 2 + 7i$ Group the real and imaginary components: - Real part: $5 - 2 = 3$ - Imaginary part: $-3i + 7i = 4i$ Combining these gives $3 + 4i$. Common misconception checks: - Choice B ($3 - 10i$) occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating $-3i - 7i = -10i$. - Choice.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

The function f is defined by $f(x) = \frac{3x+2}{x-4}$ for all real numbers $xeq4$. What is the value of $f^{-1}(2)$?

a. -10 ✓

b. -6

c. -4

d. 6

By definition of the inverse function, finding $k = f^{-1}(2)$ is equivalent to solving $f(k) = 2$. Set up the equation: $\frac{3k+2}{k-4} = 2$ Multiply both sides by $k-4$ (where $keq4$): $3k+2 = 2(k-4)$ $3k+2 = 2k-8$ Subtract $2k$ from both sides: $k+2 = -8$ Subtract 2 from both sides: $k = -10$ Thus, $f^{-1}(2) = -10$. Common misconceptions: - Evaluating $f(2) = \frac{3(2)+2}{2-4} = \frac{8}{-2} = -4$ confuses the function f with its inverse f^{-1}. - A sign error when solving.

39 [Challenging] • Target: Powers of i • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For the imaginary unit i such that $i^2 = -1$, what is the product $(4-3i)(2+5i)$?

a. $23+14i$ ✓

b. $-7+14i$

c. $23-14i$

d. $8-15i$

To find the product of the two complex numbers, expand using the distributive property (FOIL): $(4-3i)(2+5i) = 4(2)+4(5i)-3i(2)-3i(5i)$ Multiply the individual terms: $= 8+20i-6i-15i^2$ Substitute $i^2 = -1$: $= 8+14i-15(-1) = 8+14i+15 = 23+14i$ Common errors: - $-7+14i$ results from forgetting that $i^2 = -1$ changes the sign of $-15i^2$, incorrectly computing $8-15 = -7$. - $8-15i$ results from only multiplying the first terms and last terms, omitting the cross terms.

40 [Challenging] • Target: Addition and subtraction • Cross-Lesson: 6.01 Limits

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equivalent to $(4+3i)(2-5i)$?

a. $23-14i$ ✓

b. $23+14i$

c. $-7-14i$

d. $-7+14i$

To multiply the two complex numbers, apply the distributive property (FOIL): $(4+3i)(2-5i) = 4(2)+4(-5i)+3i(2)+3i(-5i)$ Compute each term: $= 8-20i+6i-15i^2$ Since $i^2 = -1$, substitute $-15i^2 = -15(-1) = 15$: $= 8-14i+15 = (8+15)-14i = 23-14i$ A common error is setting $i^2 = 1$, which incorrectly yields $8-15-14i = -7-14i$.

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41 [Challenging] • Target: Multiplication • Cross-Lesson: 2.01 Matrices & Vectors

Find all real roots of $x^3 + 8x^2 + 19x + 12 = 0$.

a. **x=-4,\ -3,\ -1 ✓**

b. x=-4,\ -3,\ 0

c. x=-4

d. x=-4,\ -3

Testing small integers with the Factor Theorem finds roots $x = -4,\ -3,\ -1$, so $f(x) = (x + 4)(x + 3)(x + 1)$ and these are all three real roots.

42 [Challenging] • Target: Conjugate and modulus • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 2x + 5 = 0$. Which root has positive imaginary part?

a. $-1 + 2i$ ✓

b. $1 + 2i$

c. $-1 - 2i$

d. $2 + i$

Discriminant $= 4 - 20 = -16$. $x = \frac{-2 \pm 4i}{2} = -1 \pm 2i$; the root with positive imaginary part is $-1 + 2i$.

43 [Challenging] • Target: Division • Cross-Lesson: 4.01 Advanced Functions & Equations

Is $f(x) = |x|$ one-to-one over all real numbers?

a. **No ✓**

b. Yes

c. Cannot be determined

d. Only for $x < 0$

A function is one-to-one iff no horizontal line crosses its graph twice.

44 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the imaginary part of $z = 6 - 7i$?

a. -7 ✓

b. 6

c. 7

d. -6

The imaginary part is the real coefficient of i, which is -7.

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45 [Challenging] • Target: Powers of i • Cross-Lesson: 6.01 Limits

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

a. $[-4, \infty)$ ✓

b. $[1, \infty)$

c. $(-\infty, -4]$

d. $(-\infty, \infty)$

1. **Identify the vertex and direction of opening**: The function $f(x) = (x - 1)^2 - 4$ is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$. 2. **Determine the minimum output value**: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4. 3. **Determine the range**: The range represents all possible.

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